

Yvahn Okmen

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ABOUT ME

Physics engineer with a strong foundation in nuclear and quantum physics, seeking to leverage academic knowledge and practical experience in a challenging and innovative environment. Looking to contribute to impactful projects that advance science and technology.

EDUCATION

MSc in Physics Engineering, Quantum Applications specialization 2022-2025

École polytechnique de Bruxelles, ULB, Brussels, Belgium

Master's Thesis – 5-month Research Internship at IBA

- **Title:** "Central region design study for stripping injection into an innovative high-intensity He^{2+} cyclotron"
- **Application:** Cancer treatment via Targeted Alpha Therapy (TAT), an in-development technique — enabled by producing ^{211}At through a 30 MeV α -beam directed onto a bismuth target.
- **Goal:** A theoretical $15\text{--}20\times$ gain in beam intensity over direct He^{2+} injection — achieved by a novel scheme that extracts the abundant He^+ from the ion source and strips it to He^{2+} inside the cyclotron (proof-of-concept).
- **Methodology:** Designed the central region's electric field map, built through three successive refinements: 2D hard-edge gap $\rightarrow$ 2D Gaussian gap (coupled with an accurate particle tracking code) $\rightarrow$ full 3D finite-difference model in Opera, solving Laplace's equation with Tosca.

BSc in Engineering / Physics Engineering 2018-2022

École polytechnique de Bruxelles, ULB, Brussels, Belgium

High school, Latin/Mathematics specialization 2012-2018

Institut Saint-Julien Parnasse, Brussels, Belgium

EXPERIENCE

Research Internship at Engie – Tractebel, PSA Team July 2023

Brussels, Belgium

- **Role:** Integrated into the Probabilistic Safety Assessment team working on advanced reactor safety.
- **Contributions:** Produced an internal state-of-the-art review of reliability assessment methodologies for passive safety systems in Gen IV reactors, covering RMPS, APSRA / APSRA+, expert-judgement approaches, Monte Carlo methods (fast-running regression surrogates and advanced MC variants), and a closing overview of dynamic PSA.
- **Deliverables:** Delivered the findings as an internal report and presented them to the PSA team.

Student Jobs 2017-2022

Brussels, Belgium

SKILLS

Key skills

- Advanced analytical reasoning
- Complex problem solving
- Research and independent learning
- Clear technical communication
- Adaptability and rigor
- Team collaboration

Computational

- MATLAB/Octave
- AOC, Opera/Tosca
- C, C++, Python
- SQL, Git

Typesetting

- $\LaTeX$
- MS Word

Languages

- French: Native
- English: Fluent
- Dutch: Basic notions (willing to improve)

PERSONAL INFORMATION

- Driving license: Category B
- Nationality: Belgian
- Date of birth: 14/11/2000